ThermoRoute: a bounded-deviation, physics-inspired framework for retrospective daily river-temperature hindcasting

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Key Points:

- A frozen protocol separates development, model freeze, predictor evidence, and one-time outcome opening
- Primary comparisons use identical station/date/horizon keys and whole-HUC2 clustered inference
- This pre-opening draft reports methods and limitations; current empirical results are intentionally absent

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Abstract

Daily river water temperature is difficult to hindcast honestly because persistence is strong, issue-time information must be separated from future observations, sensor records are incomplete, and nearby sites are not statistically independent. We describe ThermoRoute, a physics-inspired statistical predictor that combines a damped-persistence anchor, a learned thermal-relaxation proposal, a sparse horizon-conditioned variable/lag router, a causal temporal convolutional encoder, a regime mixture, a bounded point residual, separate three-quantile heads, and a split-conformal 90% interval. The canonical development data contain 657,480 site-days from 120 stable USGS site numbers during 2006–2020, spanning 34 states and 15 HUC2 groups. Years 2019–2020 are development/exploratory data. A repository-internal pre-label protocol defines a later one-time retrospective interval, six primary model classes, five formal station-balanced comparisons, exact whole-HUC2 sign-flip p-values, whole-HUC2 cluster-bootstrap intervals, and Holm multiplicity control. The protocol also freezes model/input chronology, raw outcome-quality evidence, and deterministic claim rendering. The current repository has not yet completed the canonical model rerun, model-suite freeze, predictor-evidence freeze, authorization, or opening. Accordingly, no empirical performance conclusion is made here. The contribution at this stage is a testable architecture and a fail-closed evaluation system whose remaining scientific and engineering limitations are stated explicitly.

Keywords: river water temperature; retrospective hindcasting; physics-inspired machine learning; clustered inference; conformal prediction; reproducibility.

1 Problem and scope

Water-temperature predictions may be useful for scientific monitoring and planning, but evaluation is easy to inflate. Daily temperature often changes slowly, so a copy of the last observation can be difficult to beat. Weather and flow variables can also leak future information if their timestamp or issue-time availability is not defined. Finally, rows from the same station and region are dependent; treating all daily rows as independent creates implausibly small uncertainty estimates.

This study asks a narrow empirical question: on fixed common station/date/horizon keys, can a constrained learned correction improve on strong statistical references and remain competitive with LightGBM and a global LSTM? The study separately asks whether empirical intervals and event probabilities are useful descriptions of uncertainty. It does not infer an energy budget, hydraulic state, or management utility from the architecture.

The repository contains an older three-monitoring-station case study (site IDs b1, s2, and p3) and a larger USGS development panel. The three sites are ordinary monitoring stations, not reservoirs; their legacy display order does not establish a hydraulic cascade or travel time. Those legacy outputs are not used as current evidence because they were produced before stable USGS site numbers, the 120v2 panel contract, and prediction-lineage sidecars were enforced.

1.1 Scientific context and related work

River thermal regimes reflect atmospheric forcing, discharge, groundwater, shading, channel form, and regulation; a statistical predictor does not identify that heat budget (Caissie, 2006). Earlier approaches range from nonlinear air–water regressions (Mohseni et al., 1998) to hybrid air-temperature and discharge models (Toffolon and Piccolroaz, 2015) and broader statistical comparisons (Piccolroaz et al., 2016). Recent work evaluates machine learning, deep sequence models, spatial transfer, and physics-guided river-network models (Feigl et al., 2021; Rahmani et al., 2021; Jia et al., 2021; Zwart et al., 2023). ThermoRoute's narrow contribution is a common-key, clustered, pre-opening benchmark of a bounded correction; it is not a claim to replace these process or network models.

The implementation uses USGS NWIS daily values, Daymet meteorology, and gridMET wind (USGS NWIS; Daymet V4; Abatzoglou, 2013). LightGBM is the principal tree baseline (Ke et al., 2017), and interval calibration follows conformalized quantile regression (Romano et al., 2019).

2 Data

2.1 Canonical development panel

The canonical file is `data_usgs/panel_usgs_120v2.parquet`, bound to `data_usgs/station_registry_v1.csv` by `data_usgs/frozen_panel_v1.json`. Each of 120 sites has one row for every calendar day from 2006-01-01 through 2020-12-31, giving 657,480 rows. The registry contains 34 states and 15 HUC2 groups. The raw columns consumed by the model, in frozen order, are water temperature (`WTEMP`), streamflow (`FLOW`), air temperature (`TEMP`), precipitation (`PRCP`), a relative-humidity proxy (`RHMEAN`), Daymet daylight-period mean incoming shortwave radiation in W m^{-2} (`DH`), and wind speed (`WDSP`). `DH` is not day length, and `RHMEAN` is derived from temperature and vapor pressure rather than a direct daily-mean humidity observation. Water level may be retained as raw evidence but is not a model input.

Observed-value missingness in the frozen panel is approximately 15.8% for `WTEMP`, 2.8% for `FLOW`, about 0.07% for `TEMP`, `PRCP`, `RHMEAN`, and `DH`, and zero for `WDSP`. These rates describe the committed derived panel; they do not reconstruct provider request history. The small Daymet gap is structured rather than random: all 120 sites lack the provider-calendar final day in leap years 2008, 2012, 2016, and 2020. `WLEVEL`, retained only as raw evidence, is missing on 77.65% of rows.

The pre-opening environmental audit records 1,345 rejected candidate stations in the legacy selection ledger (950 without joint WTEMP/FLOW availability, 376 with low full-period coverage, and 19 with low 2019–2020 development-period coverage). Among retained stations, 38 share a repeated HUC code with another retained station and 19 have another retained station within 10 km. WTEMP and FLOW have maximum missing runs of 2,404 and 1,369 days, respectively. Two stations contain 2,059 signed negative-FLOW rows (minimum −121 cfs), whose tidal, backwater, or measurement semantics are not resolved. These facts reinforce that the cohort is availability-enriched and spatially dependent, rather than nationally representative.

The surviving station ledgers are consistent with full-period WTEMP/FLOW coverage thresholds of 0.55/0.70 and 2019–2020 thresholds of 0.80/0.80. Consequently, 2019–2020 participated in cohort construction as well as later model and narrative development. The original discovery responses, retrieval timestamps, exact command line, and complete run configuration were not retained. The original 1,465-candidate selection is therefore auditable from committed ledgers but is not source-replayable. The development panel also retains no NWIS qualifier columns, method or sensor history, or original daily-value response bytes; measurement discontinuities and qualifier sensitivity cannot be reconstructed for 2006–2020.

The temporal roles are fixed before fitting:

Role	Dates	Rows	Observed `WTEMP`	Permitted use
Training	2006–2015	438,240	341,646	fit preprocessing and models
Validation	2016–2017	87,720	84,074	choose frozen settings
Calibration	2018	43,800	42,279	CQR and event-reference calibration
Development evaluation	2019–2020	87,720	85,621	previously inspected; exploratory diagnosis only

The 2019–2020 partition was inspected and informed cohort, model, and narrative development; it is exploratory and not independently confirmatory.

2.2 Target-period inputs and outcomes

The target interval is fixed in the protocol. Historical Daymet and gridMET covariates must be retrieved and archived before outcome access. Meteorology is represented at each station coordinate, not aggregated over upstream catchments. The primary information set uses values dated on or before each historical issue date in retrospectively retrieved, finalized products and consumes no horizon-specific future weather forecast. This is a date-indexed retrospective hindcast; it does not establish that the same predictor vintages were available operationally at that time.

Because the legacy panel lacks its original HTTP responses, a separate pre-opening bridge re-fetched 2018–2020 Daymet/gridMET with the confirmation parser and compared every predictor on the exact site/date registry. Its committed manifest records `PASS_EXACT_PRODUCT_BRIDGE`. This gate can detect product-version, parsing, scaling, missingness, and one-day alignment drift. It cannot prove that provider calendar days share NWIS local-day boundaries, nor can it reconstruct as-issued data availability; those remain permanent limitations.

After the final local protocol seal, we added this outcome-free 2018–2020 parser/product compatibility gate. The post-seal engineering gate strengthens reproducibility but was not part of the sealed protocol and does not alter its hypotheses, estimands, model registry, margins, or decision rules. The committed offline comparison has passed; later gates must retain its exact path and digest.

The raw-only opening child is restricted to USGS daily-value requests for daily mean water temperature, streamflow, and water level. It records exact request and response bytes, series identifiers, qualifiers, final URLs, timestamps, and hashes. If simultaneous finite series disagree, the normalized value is missing and every constituent is retained in the quality audit. No site can be replaced after outcomes are accessed.

2.3 Metadata-only external cohort

After the model-suite commit, a fixed 34-state metadata query identifies stream sites advertising daily water-temperature capability without requesting values or coverage dates. A deterministic seed selects 30 stations whose exact USGS site IDs do not occur in the development registry. This is site-ID disjointness, not proof of HUC8, river-network, upstream/downstream, or distance separation. Models for this cohort are pooled and station-agnostic where applicable. The exercise remains outside the five-test family and cannot change model or site selection.

3 Model design

3.1 Conservative anchor and learned proposal

Let `y_t` be the last observed daily water temperature and `c_{t+h}` a frozen seasonal climatology. Damped persistence moves `y_t` toward `c_{t+h}` at a fixed rate and provides the principal anchor. ThermoRoute also produces a flow- and season-conditioned relaxation proposal. That proposal is learned from data; its parameter is not interpreted as a measured residence time or heat-transfer coefficient.

3.2 Variable/lag router, TCN, and regime mixture

For each horizon, a sparse router assigns weights over the seven variables at lags 0–14. The sequence builder supplies a 32-day tensor, but this is a construction buffer rather than an effective-memory claim: the current two-block, kernel-three causal temporal convolutional network has a theoretical seven-step receptive field, while the router's oldest usable value is lag 14. Thus no input older than lag 14 can affect the current model output. A mixture-of-experts combines the resulting representations. The router is an allocation mechanism inside the predictor, not a map of connected reaches. These components follow sparsemax attention (Martins and Astudillo, 2016), generic temporal convolutional sequence modeling (Bai et al., 2018), and sparsely gated expert mixtures (Shazeer et al., 2017).

3.3 Bounded residual and probabilistic heads

The learned point correction is transformed by `tanh` and scaled by a fixed `delta_scale`, limiting its numerical deviation from the anchor. This yields a finite algebraic bound relative to that reference for the point output. It does not bound absolute error, event-tail error, interval width, or behavior after a distribution shift.

The learned models emit a separate MSE point forecast and pinball-trained 0.05, 0.50, and 0.95 quantile heads. Split conformalized quantile regression fitted on 2018 widens only the nominal q05–q95 interval; q50 is not CQR-adjusted. Ensemble quantiles are member-wise averages, not quantiles of a mixture distribution. Neural heads are ordered by construction, making the retained crossing-loss term identically zero and compatibility-only. Independently fitted LightGBM heads are not sorted: raw development crossings are recorded by member and horizon, q05 and q95 are clipped to the nominal raw q50 if needed, and q50 is left exactly unchanged. Temporal CQR offsets are station-by-horizon, whereas external offsets are pooled by horizon. A separate event head uses each temporal station's 2006–2015 q90, or one pooled development-training q90 in the external arm, and one Platt calibrator per horizon fitted on 2018. Platt fitting is calibration-row weighted, whereas target-period probability summaries give every retained station equal total weight; that frozen weighting difference is part of the interpretation. The summaries claim empirical marginal coverage only. Quantile crossing, station-balanced achieved coverage, interval width, three-quantile pinball mean, Brier score, log loss, discrimination, reliability, and calibration slope/intercept are descriptive outputs.

4 Evaluation design

4.1 Primary models and common keys

The six primary model types are persistence, damped persistence, climatology, LightGBM, a global LSTM, and ThermoRoute. The three deterministic references have no seed; LightGBM, LSTM, and ThermoRoute use seeds 0–4. LightGBM selects among four candidate settings separately by horizon on 2016–2017; the LSTM selects among three candidate architectures with seed 0 before fitting five members. These tuning budgets are documented but not identical. The temporal mechanism audit additionally freezes seven one-factor controls: prior-only, no dynamic prior, fixed relaxation, no router, no mixture, no TCN, and unbounded residual. Each Stage 09 control is a seed-0-versus-seed-0, single-seed functionality/intervention diagnostic. It does not prove component necessity, identify a causal mechanism, or establish cross-seed stability. Every comparison uses identical station/date/horizon target keys for the two models.

The temporal arm is future-time hindcasting at known gauges. ThermoRoute and the LSTM receive a stable station embedding, and temporal LightGBM receives the same site identity as a categorical feature. The external pooled arm disables all station-identity inputs and uses only pooled development-training transformations, but still consumes each target site's observed WTEMP history through the issue date. It is therefore a history-dependent new-gage evaluation and cannot be interpreted as prediction without target-site water-temperature records.

An admissible issue must have genuinely observed issue-date WTEMP. Other cells in the 32-day history may be filled with training-only seasonal medians and retain explicit missingness masks; no minimum observed-history fraction is required. Development windows require all three target horizons to remain within one partition and have observed labels, whereas later confirmation evaluates horizon admissibility independently. FLOW uses signed `log1p`, preserving negative values rather than clipping them or calling them drought; PRCP uses nonnegative `log1p`. Temporal scaling and imputation are station-specific and train-only, while the external arm uses pooled train-only transformations.

The current canonical configuration sets `delta_scale` to 1.0 °C. It is a development-selected algebraic point bound relative to the damped anchor, not an error, tail-risk, or safety bound. Earlier 2019–2020 inspection informed project development, so this setting is not presented as selection from an untouched development test.

For each station and horizon, RMSE is calculated only when at least 100 paired targets are valid for both models. The primary effect is the unweighted median of station-level candidate-minus-reference RMSE. Daily rows therefore do not directly determine the between-station weight.

4.2 Formal five-test family

The frozen family contains:

1. ThermoRoute versus damped persistence at 1 day, margin 0.00 °C.
2. ThermoRoute versus damped persistence at 3 days, margin 0.00 °C.
3. ThermoRoute versus damped persistence at 7 days, margin 0.00 °C.
4. ThermoRoute versus LightGBM at 3 days, numerical ceiling +0.05 °C.
5. ThermoRoute versus LightGBM at 7 days, numerical ceiling +0.05 °C.

Each raw one-sided p-value is obtained by enumerating every whole-HUC2 sign configuration for the frozen reportable cohort. A 10,000-draw whole-HUC2 cluster bootstrap provides the interval for the median station effect. Holm adjustment is applied to exactly these five p-values (Holm, 1979). A favorable formal statement is allowed only when the row is estimable, the Holm p-value is at most 0.05, and the interval upper bound is strictly below the frozen margin. Disagreement between the p-value and interval rules is reported conservatively. The 1-day LightGBM comparison is descriptive only.

The H2 claim is scoped only to the frozen LightGBM procedure: four predeclared candidate settings are compared on 2016–2017 and the selected setting is then fit with the frozen five seeds. H2 is not a claim about LightGBM generally, a larger hyperparameter search, or a differently resourced implementation. Historical tuning budgets were not equalized across model classes.

The sign-flip procedure assumes independent whole-cluster sign reversals and joint sign symmetry of each complete HUC2 effect vector around the tested margin. Exact enumeration removes Monte Carlo error but does not make this assumption distribution-free. Before reportability attrition, the 15 HUC2 groups contain 2–26 stations, the largest contains 21.7% of stations, and the inverse-Herfindahl effective cluster count is 9.54. The percentile cluster-bootstrap interval is therefore interpreted cautiously. Effective-cluster, largest-share, and leave-one-HUC diagnostics carry NO_STRONG_INFERENCE when their predeclared small/unequal-cluster warnings fire; they do not change the sealed five-test rule.

4.3 Secondary analyses

Predeclared secondary outputs include exact-approved-qualifier target sensitivity, equal-HUC and leave-one-HUC influence summaries, station-balanced probability diagnostics, architecture controls, and synthetic missingness/noise/flow/weather perturbations. They cannot promote or replace a failed formal test. A hypothetical cost–loss calculation, if retained, is an illustration rather than observed decision impact.

The optional Air2stream-style a4/a8 implementation is an unofficial, exploratory style reference fitted by deterministic six-start bounded least squares on measured drivers. It is not the official Air2stream implementation or a validated reproduction, and no result against it supports a claim against the published model. A published-model comparison requires the official implementation and documented calibration search.

4.4 Temporal coverage and calendar-balance audit

The predeclared audit is restricted to the fixed, availability-enriched Route-A cohort and forecast keys for which both issue-date and target-date WTEMP are retained finite observations in the normalized outcome tables. It does not impute unavailable outcomes, test missing at random, estimate all-calendar-day performance, or establish stability across years or seasons.

For every formal comparison, the audit reports all eight frozen descriptive candidates: equal weighting of the 12 year-by-season cells, leave-one-year for each of 2021–2023, and leave-one-season for DJF, MAM, JJA, and SON. The maximum candidate-minus-reference effect is the deterministic unfavorable value, with exact ties resolved by the frozen candidate order. The audit never filters or replaces a primary station and cannot alter the formal effect, p-value, confidence interval, Holm decision, or claim eligibility; a favorable sensitivity cannot rescue the primary result. If a formal test is NOT_ESTIMABLE, its formal effect is NA, while any separately computable prediction-derived effect remains explicitly descriptive and cannot upgrade inference. The artifact becomes evidence only after independent physical replay of every bound source and exact binding into the opening receipt and completed release.

5 Reproducibility and one-time opening

The final protocol bytes are bound to both their original Git preregistration commit and the final pre-label commit by SHA-256. The evidence explicitly lacks an external timestamp or independent custodian. The claim ledger derives phase only from the canonical authorization, intent marker, and a fully verified receipt.

Before authorization, a fresh isolated process must replay all trained members, heads, transforms, development predictions, dependency versions, runtime probe, and source hashes without reading target-period outcomes. Git chronology must show that the model-suite commit predates candidate metadata and target-period predictor artifacts. Any source change after the model freeze invalidates authorization.

The opening creates one irreversible intent and one fixed request ledger. This is one logical opening, not exactly-once HTTP delivery. Before the acquisition manifest exists, a network continuation may fetch only ledger entries without a durably published canonical transaction, under the same opening identifier, and only while normalized, derived, and trusted outputs are absent. HTTP retries are allowed; a response received before its transaction directory is complete and durable may be requested again, whereas a complete, durable, verifiable canonical response is never replaced. A partial, invalid, or noncanonical canonical transaction fails closed without overwrite; cleanup is limited to unpublished owner-private temporary or pending state. Once all raw transactions validate, the request map, two normalized tables, and manifest are generated and validated in one private same-filesystem stage and published by one directory rename. After the manifest exists, raw-network continuation and relaunch of the raw child are prohibited, but network-free deterministic completion under the same opening identifier remains allowed. The scorer builds and validates one complete private same-filesystem trusted directory before one directory-rename publication. An absent canonical generation is recomputed; a complete canonical generation without a receipt is fully replayed before receipt creation.

An abandoned stage is deleted only after its canonical name, owner-private mode, same-device read-only regular files, and absence of external hard links all validate; any unsafe stage fails closed. Partial, invalid, or noncanonical canonical contents fail closed and are not replaced. A valid receipt missing only its external digest sidecar may regenerate that sidecar after full receipt-and-artifact validation; a sidecar without a receipt fails closed. This is an honest-owner crash/replay guard, not protection from a malicious owner or same-UID adversary.

6 Current progress

Completed repository work includes the canonical panel/registry seal, the final pre-label protocol, deterministic model serialization contracts, development replay code, exact clustered inference, probability/event contracts, outcome-quality provenance, the committed `PASS_EXACT_PRODUCT_BRIDGE` evidence, the predeclared temporal-coverage audit and its receipt-binding implementation, a dual-profile release verifier, and a structured claim ledger.

Required work still outstanding at this manuscript state is:

1. rerun the canonical six-model development experiment, the seven mandatory seed-0-versus-seed-0 diagnostic controls, and the separate matched plain-neural/feature-ladder audit;
2. freeze and independently replay every five-member temporal and pooled external learned-model bundle;
3. commit the model-suite registry before obtaining external metadata and target-period predictor artifacts;
4. freeze those outcome-free inputs and validate the Git chronology receipt;
5. run the full preflight suite, including the implemented same-opening transport continuation, before creating the unique authorization;
6. execute the fixed opening, physically replay and receipt-bind the temporal-coverage audit, render all five receipt-derived statements, regenerate artifacts, and verify the clean-room release.

Until those steps finish, all legacy numeric outputs are considered stale and no current empirical performance conclusion is available.

7 Permanent limitations

The following machine-verifiable statements are part of the evidence contract and remain in the manuscript regardless of the eventual numerical outcome.

Route A is history-dependent and uses target-site water-temperature observations through each issue date; it does not establish ungauged prediction.

Route A does not establish independent river-network separation, physical river-network routing, or hydraulic travel time.

Route A is a one-shot retrospective historical-information evaluation, not an operational replay with archived as-issued predictor vintages or future NWP.

Route A uses daily-mean statistical thresholds and a numerical non-inferiority margin; neither has ecological, biological, or regulatory meaning.

Route A provides no physical, deployment, regulatory, or distribution-free safety guarantee, and failure to reject is not equivalence.

Route-A architecture controls and predictor sensitivities are descriptive or exploratory and do not identify causal mechanisms.

Route A evaluates a fixed availability-enriched 120-site cohort and is not nationally representative of all U.S. rivers or all calendar days.

Route A does not establish conditional coverage, and its equal-weight three-quantile pinball summary is not CRPS.

The optional ecological-threshold stage is separate from prediction evaluation. It can only compare independently sourced observed daily maxima, expressed as a strict seven-consecutive-day 7DADM, with a complete site-specific standards registry. Seasonal applicability is assigned by the seven-day window's ending date; season-external or incomplete observation windows remain unclassified. Missing, unrelated, or ambiguous standard matches fail closed. Any reported observed threshold exceedance is descriptive and is not a model result or a legal decision; it does not determine compliance with any law or regulation.

Additional limitations are the availability-enriched station sample; missing original provider bytes for the development panel; point-scale meteorology rather than upstream forcings; unbalanced and coarse HUC2 clusters; daily-mean outcomes; outcome-observability conditioning; a numerical margin without stakeholder-derived importance; retrospective finalized covariates; absence of latency, memory, energy, and multi-hardware benchmarks; owner-controlled local evidence chronology; the unreplayable original 1,465-candidate discovery execution; and the absence of an official Air2stream calibration run.

8 Pre-opening results placeholder and stopping rule

No formal result block is present in this byte-frozen pre-opening snapshot. After a valid receipt exists, `python scripts/26_validate_claims.py --root . --registry protocols/route_a_claim_registry_v1.json --write-generated-results --require-complete` derives five deterministic blocks from the exact statistics artifact and atomically and idempotently writes the sole canonical post-opening result layer. Each must appear exactly once, including negative, conflicting, or non-estimable outcomes. Handwritten substitutions are rejected by body hashes and evidence bindings. Once that generated layer exists, the readiness statements in this snapshot are historical provenance rather than the current execution status.

9 Data and code availability

The canonical derived panel, stable station registry, HUC metadata snapshot, code, protocol, and tests are in this repository. Original HTTP bytes and retrieval timestamps for the 2006–2020 panel are unavailable, so replay begins from the committed Parquet artifact. Candidate, meteorological, and target-outcome requests are designed to retain exact raw evidence in a completed release. A DOI, public repository URL, data license, and verified author metadata have not yet been assigned and must be completed before submission.

10 Conclusion

At the time of this byte-frozen pre-opening snapshot, ThermoRoute is a specified research system rather than a completed empirical finding. The repository has fail-closed controls for leakage, cohort identity, model replay, clustered inference, raw provenance, statement generation, and release closure. Its scientific value will depend on a complete canonical rerun and the one-time evidence chain. The implementation must not enter confirmatory opening until every listed data, bridge, replay, chronology, authorization, and release gate passes. If a generated receipt-derived result layer appears below, it supersedes only this pre-opening readiness status; the old headline numbers still do not represent the current repository.